

Aare Renu Shri

B.Tech Computer Science and Engineering, IIT Hyderabad · Expected 2028

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EDUCATION

IIT Hyderabad — B.Tech in Computer Science and Engineering **Expected 2028**

CGPA: **7.3 / 10** | JEE Advanced 2024: **AIR 3347** | JEE Main 2024: **AIR 3336**

Sri Chaitanya Junior Girls College, Madhapur, Hyderabad — XII, Telangana Board **98.40%**

Sri Chaitanya Tech HS, Sai Mourya Towers, Kodad — X, Telangana Board **10.00 CGPA**

Relevant Coursework: Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Architecture, Compilers, Software Development Fundamentals, Discrete Mathematics, Probability & Random Variables, Linear Algebra

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, RISC-V Assembly

Frameworks/Tools: FastAPI, Docker, Render, Git, GitHub, QEMU, gem5

Systems: xv6 / UNIX-like OS internals, Linux/Unix (Terminal)

Other: Flex (lexer generation), LaTeX, Digital Fabrication Tools

PROJECTS

Log Classifier API | FastAPI, Docker, Render | Personal Project **Live Demo** | **GitHub**

- Developed an **NLP pipeline** classifying log messages into **4 categories** using **TF-IDF + Logistic Regression**, achieving **99.6% accuracy** on a held-out test set of 248 logs.
- Designed **regex-based log normalization** to replace variable values with placeholder tokens, fixing a **train/inference preprocessing mismatch**; validated the model on **32 unseen logs**, achieving **96.9% accuracy**.
- Exposed the trained model through a **FastAPI /classify endpoint**, using **Pydantic** for request validation and **Swagger UI** for interactive endpoint testing.
- Configured **Render deployment** with custom build/start commands, **port binding via environment variables**, and host configuration for external traffic, making the API publicly accessible.

Query Optimization & Indexing Analysis | PostgreSQL, Python **GitHub**

- Benchmarked indexing strategies on a **6M-row TPC-H dataset**, measuring **~18% faster load times** when creating an index after bulk data insertion versus before, using isolated table copies and controlled timing experiments.
- Built a Python benchmarking tool measuring query performance across **5 indexing strategies** (no index, index, forced index, clustered index, index-only scan) over **50 selectivity levels**, using **warm-cache** methodology to eliminate disk I/O noise and ensure consistent comparisons.
- Implemented a recursive **query-plan comparator** in Python that parses PostgreSQL's **EXPLAIN (FORMAT JSON)** output and determines structural equivalence between plans, ignoring cost and cardinality noise.
- Designed a **divide-and-conquer algorithm** to detect optimizer plan switches across a parameterized date range, reducing required queries by **~95% (16 vs. ~365)** versus brute-force search, with results independently verified.

xv6 OS Kernel Extensions | C | Course Project (OS II) **GitHub**

- Added new **system calls** in xv6 for processes to query their own info, backed by **kernel-side syscall-count tracking**.
- Engineered a **4-level priority scheduler (4-level-SC-MLFQ)** using syscall-count tracking to classify **interactive vs. CPU-intensive** processes, with aging to prevent starvation.
- Implemented **virtual memory** using a **clock page replacement algorithm**, allowing xv6 to swap pages to disk instead of terminating processes when physical memory is full.
- Extended virtual memory with **disk-backed swapping**, added **FCFS and SSTF disk scheduling**, and integrated **RAID 0/1/5** to improve storage performance, reliability, and fault tolerance across simulated disks.

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RISC-V FPU vs Load/Store Performance | RISC-V Assembly | **GitHub**
Course Project

- Implemented and benchmarked **baseline load/store-based** and **FPU-optimized matrix multiplication** routines in RISC-V assembly using **gem5**, evaluating performance across matrix sizes from **16×16 to 128×128**.
- Optimized matrix multiplication using **FPU register-based operations** to reduce memory accesses, achieving an approximately **6-7% improvement in execution time** over the load/store-based baseline.
- Analyzed **gem5 simulation results** across matrix sizes to compare the execution performance of baseline and FPU-optimized implementations.

Log Analyzer | Python | Personal Project

[GitHub](#)

- Built a Python CLI tool that parses log files using **compiled regex patterns**, safely skipping malformed or corrupted entries instead of crashing or silently miscounting them.
- Used Python's **Counter** to analyze error frequency across **500+ log lines**, identifying the most common failure reasons and tracking daily error trends across datasets.
- Generated **daily error trend** and **error distribution** visualizations with Matplotlib, converting raw log data into clear visual summaries.
- Built a **synthetic log generator** using weighted random sampling and randomized timestamps to produce realistic multi-day test data for systematic validation of parsing and analysis logic.

POSITIONS OF RESPONSIBILITY

Event Coordinator — DIESTA, IIT Hyderabad

2nd Sem, 2025

- Monitored squash matches across multiple rounds, coordinating with department representatives to resolve scheduling conflicts and logistical issues.

Coordinator, Cultural & Design — CSE Departmental Day, IIT Hyderabad

2nd Sem, 2025

- Contacted department alumni across graduating batches and coordinated an interactive alumni-student discussion session.

Deputy Coordinator — Elan & nVision, IIT Hyderabad

2nd Sem, 2025

- Publicity: Contacted colleges outside IIT Hyderabad to promote the fest and increase student participation. Informals: Helped construct a large-scale cardboard showcase installation.

EXTRACURRICULAR

- NSS (National Service Scheme), IIT Hyderabad — Contributed to campus cleanliness drives, walkathons, and awareness campaigns on social and environmental causes.
- Hobbies — Dance, Sketching and Arts.